

Analysis note 5.0.1

Agentic guidelines:

Be efficient with token usage. In addition to the managing agent, a figure design specialist agent could be good to have on retainer here as many of my comments that follow requests additional informational displays

Make the following changes with similar initial instructions being kept in mind:

- Figures 57, 58, 59, 60 can all be combined into a single figure that has 2015 top left, 2016 top right, 2021 10% bottom left and combined correlation matrix bottom right.
- It appears that a series of plots from the 4.9.7 analysis note were not propagated to the 5.0.0 document
 - Figure 8, 9, 10, 11, 12 and corresponding captions need to be placed in section 3 and a new subsection summarizing this series of kinematic distributions figures should be made.
- Clean up figure 21 to have less overlapping text in the primary part of the figures
- In this section "Resolution-scaled length-scale parameterization and bounds" there used to be a nice/illustrative figure that showed the allowed length scale regions for each dataset in GP space..
- After the line "with $L_d(m)$ the lower-triangular Cholesky factor of the positive-semidefinite blind-window covariance." Could be a good place to include a subsection that is "profiled background implementation" that write out formally what the math is for the profiling, and then mention how it is used with some illustrative diagram
- At the beginning of 4.8, a three or four block flow chart would be nice to summarize this extraction of signal and conversion to ϵ_2
- At the end of section 4.9 I would like reference to the new global significance calculation technique outlined in section 4.19
- In section 4.10 is the line "The second line is the Wald branch" — this is confusing, what is Wald mean
- In section 4.13, I don't like the equation centering. At the beginning of this sub-section, I would also like a small flow chart showing how the shared ϵ_2 is constructed physically.
- Section 4.19 should have a lot of the equations more cleanly written out so it isn't so cluttered
 - After the line "then generates many inexpensive correlated fields." " should probably be an informative diagram or selection from their

paper so it is clear to the HPS physicists what exactly we gain from this new method of significance calculation.

- Section 5 should just be background model validation with pseudo-experiments
- Figure 29 should probably be placed at the beginning of section 5
- In section 5.1 after the line "We use four exposure ensembles." There could be a bulleted list, it should be specified that the 100% statistics are used to determine what model should be fixed for the final unblinding.
- Section 5.2 should be probably renamed into "optimizing blind window" and there should be a historic plot that shows more broadly the different validation plots that assisted in making this decision
- In section 5.5 after this line "The correlation calculation is a fixed-hyperparameter statement" you could mention section 6.9 to that the correlations might imply on the signal model extraction
- The section 5.10 of 4.9.7 had a really useful table that should probably be placed at the beginning of this section slightly edited to make it more clear what figures tested what study and what the decisions made ended up being
- In section 6.9 all of the figures that are made need to be explicitly referenced and the results should be summarized.
- Figure 62 should state what the first and second observed scan regions that were replaced are.
- The figure 74 controlled 2016 upper range study should be included in the current appendix as it gave us the decision on the upper length bound
- We should have a version of the 4.9.7 figure 84 that shows current and projected reach compared to BaBar
- A version of the 4.9.7 figure 85 should be made for each of the three datasets and included in the independent dataset results section
- The B.4. three toy generation section should be added to the 5.0.1 appendix and then the exact functional forms used to generate the pseudoexperiments should be written out explicitly in the generating spectra section.
- B.6 from 4.9.7 should be included in appendix as well There should be a small subsection or mention dedicated to the threshold modeling that we used to determine GP validation at 65 MeV (as this is included in the primary validation plots now)
- Section E of 4.9.7 note should be a section in the appendix, additionally I believe I made a comparison plot of 2016 95% GPR to the previously published result, so I would like that also to be in the appendix, perhaps the two sections could be merged as a comparison of previously analyzed data. The 2016 approach should also be detailed a bit more

explicitly.